SEC:: SR. JEE MAINS APEX | GTM-17 | Date: 11.01.2025
Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
A	D	A	A	A	C	C	B	A	B
11	12	13	14	15	16	17	18	19	20
A	B	D	B	D	D	C	A	B	B
21	22	23	24	25					
17	200	20	36	25					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
C	B	D	D	C	A	D	A	B	B
36	37	38	39	40	41	42	43	44	45
A	C	D	C	C	B	C	A	A	D
46	47	48	49	50					
3	3	4	2	4					

MATHEMATICS

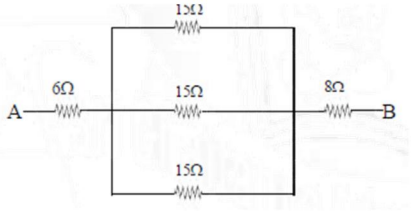
51	52	53	54	55	56	57	58	59	60
D	D	D	A	B	C	C	C	C	D
61	62	63	64	65	66	67	68	69	70
A	D	A	D	D	D	B	B	C	A
71	72	73	74	75					
25	81	9	14	19					

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Hints&sol

PHYSICS

1.



$$R_{Am} = 14 + \frac{15}{3} = 19\Omega$$

2. $n = 1, v_1 = v, v_2 = 2v$

$$r = \frac{3}{2}, w = ?$$

$$\left[w = \frac{nR}{r-1}[T_1 - T_2] \right] \text{ but } T_1 v_1^{r-1} = T_2 V_2^{r-1}$$

$$T_2 = T\left[\frac{v}{2v}\right]^{\frac{3}{2}-1}$$

$$T_2 = T \times \frac{1}{\sqrt{2}}$$

$$\therefore w = \frac{R}{1/2}\left[T - \frac{T}{\sqrt{2}}\right] \Rightarrow w = 2RT\left[\frac{\sqrt{2}-1}{\sqrt{2}}\right] = \sqrt{2}RT[\sqrt{2}-1]$$

$$w = RT[2 - \sqrt{2}]J$$

3. Concave lens always forms vertical and erect image

4. $L = \frac{Q}{m} \Rightarrow L.H = L^2T^{-2}$

5. Conceptual

6. $F_B = mg$

$$P_w \times v_1 \times g = p_s vol \times g$$

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$$\Rightarrow \pi \times \frac{4}{3}\pi\left[\frac{D^3}{8}-\frac{d^3}{8}\right]=3\times\frac{4}{8}\pi\frac{D^3}{8}$$

$$\Rightarrow \pi\sigma(D^3-d^3)=\frac{\pi}{6}D^3$$

$$\Rightarrow \frac{1}{\sigma}=1-\frac{d^3}{D^3}$$

$$\Rightarrow \frac{d^3}{D^3}=\frac{\sigma-1}{\sigma}$$

$$\Rightarrow \left(\frac{D}{d}\right)^3=\frac{\sigma}{\sigma-p}\Rightarrow\frac{D}{d}=\left(\frac{\sigma}{\sigma-1}\right)^{\frac{1}{3}}$$

7. $1MSD=M$

$nMSD=(n+1)VSD$

$$LC=1MSD-WSD=MSD-\left(\frac{n}{n+1}\right)MSD$$

$$LC=\frac{m}{n+1}$$

8. $$C_1=\frac{A/3\in_0}{d};c_2=\frac{A/3\in_0}{2d}:c_{est}=c_1+c_2+c_3=\frac{11A\varepsilon_0}{18d}$$

9. T.E on the surface T.E$=E_1=\frac{gmm}{2R}$

T.E in the orbit $=E_2=\frac{-Gmm}{2(r+h)}$

$$\Delta E=\frac{Gmm}{2R}-\frac{Gmm}{2(R+h)}=\frac{Gmm}{2}\left[\frac{1}{R}-\frac{1}{R+h}\right]=\frac{Gmm\times 20.4}{2\times 21h}$$

10. $V=\alpha\sqrt{x}$

At $x=0;v_1=0$

$x=d;v_2=d\sqrt{d}$

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$$w = \Delta k = \frac{1}{2} m\left[v_2^{\,2} - v_1^{\,2}\right] = \frac{m\alpha^2 d}{2}$$

11. G=200Ω, $I = 20\mu A$

$$I = \frac{G}{\frac{i}{I_g - 1}} = \frac{200}{1000 - 1}$$

$$S = \frac{200}{999}\Omega = 0.20\Omega$$

12. $$a = \frac{m_1 - m_2}{m_1 + m_2}$$

$$\frac{g}{8} = \frac{m_1 - m_2}{m_1 + m/2} g$$

$$8m_1 - 8m_2 = m_1 + m_2$$

$$7m_1 = 9m_2$$

$$\frac{m_1}{m_2} = \frac{9}{7}$$

13. $$\lambda = \frac{h}{\sqrt{2mk}}, \lambda\alpha \frac{1}{\sqrt{m}}$$

As $m_e < m_p < m_\alpha \Rightarrow \lambda_e > \lambda_p > \lambda_\alpha$

14. $p = \frac{v^2}{x_c} = v^2(wc)$ as dielectric material is introduced capacity increase hence power increase growing of the bulb increases

15. m. field lies along Z-axis

$$B_2 = B_0 \,\mathrm{Sin}[kx - wt]$$

$$B_0 = \frac{\varepsilon_0}{c} = 2 \times 10^{-7} t$$

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$$k=\frac{2\pi}{\lambda}=\frac{2\pi}{4}\times 10^3=\frac{\pi}{2}\lambda 10^8$$

$$\therefore B=2\times 10^{-7}\sin\left[\frac{\pi}{2}\times 10^3[x-3\times 10^8]\right]\rightarrow \hat{k}T$$

16. $\sqrt{a^2+b^2+2ab\cos\theta}=\sqrt{2}[\sqrt{a^2+b^2-2ab\cos\theta}]$

$$\Rightarrow a^2+b^2+2ab\times\frac{5}{9}=2\left(a^2+b^2-2ab\times\frac{5}{9}\right)$$

$$\Rightarrow a^2+b^2=6ab\cos\theta \qquad \Rightarrow a^2+b^2=6ab\times\frac{5}{9}$$

$$\Rightarrow n^2+b^2=6nb\times\frac{5}{9} \qquad \Rightarrow 9n^2-30n+9=0$$

17. $E_g=\frac{hc}{\lambda}\Rightarrow 1.42ev=\frac{1240ev-nm}{\lambda}$

$$\Rightarrow \lambda=\frac{1240}{1.42}nm\approx 875m$$

18. For half of distance $V_AV_B=\frac{2V_1+V_2}{V_1+V_2}$

$$V_1=6m/s$$

$$V_2=\frac{9+15}{2}=12m/s$$

19. $p_1v_1^{\ 2}=p_2V_2^{\ 2}$

$$\Rightarrow\frac{p_1}{p_2}=\left[\frac{v_2}{v_1}\right]^2$$

$$\Rightarrow\frac{p_1}{p_2}=\left[\frac{4}{5}\right]^{3/2}$$

$$=\frac{8}{5\sqrt{5}}$$

20. $e=mc^2J$

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$$10^{-3}\times(3\times108)^2 J = 9\times10^3 J = \frac{9\times10^{13}}{1.6\times10^{-13}} MeV$$

$$\Rightarrow E = 5.6\times10^{26} MeV$$

21. $a = 16 m/s^2$

$$\Rightarrow w^2 4 = 16$$

$$\Rightarrow w^2\times4 = 16$$

$$\Rightarrow w = 2 rad/s$$

$$V = w\sqrt{A^2-4^2}$$

$$\Rightarrow 2 = 2\sqrt{A^2-16} \Rightarrow A = \sqrt{17} m$$

22. $I = I_{\frac{\max}{4}} \Rightarrow I_{\max}\cos^2\frac{\theta}{2} = \frac{I_{\max}}{4}$

$$\Rightarrow cso\frac{\theta}{2} = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{2\pi}{3}$$

$$\Rightarrow \theta = \frac{2\pi}{\lambda}(\Delta x)$$

$$(\Delta x) = \frac{\pi}{2\pi}(\pi) = \frac{\lambda}{2\pi}(\frac{2\pi}{3}) = \frac{\lambda}{3}$$

$$y = \frac{\lambda D}{3d} = 200 nm$$

23. $\Delta l = \frac{Fl}{Ay} \Rightarrow \Delta l = \frac{200\times2}{2\times10^{-4}\times10^{11}} = 2\times10^{-5} m$

24. Potential $v = \frac{\pi\lambda}{4\pi\in_0} = \frac{4\times10^{-9}}{4\times885\times10^{-12}}$

$$\Rightarrow v = \frac{\pi\lambda}{4\pi\in_0}$$

$$= \pi\times4\times10^{-9}\times9\times10^9 V = 36\pi v$$

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25. KCL

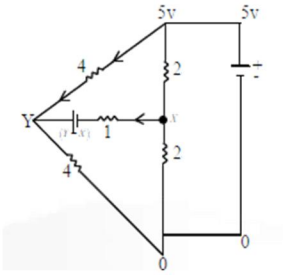
$$\frac{x-5}{2}+\frac{x}{2}+\frac{x-y+10}{1}=0$$

$$\Rightarrow 4x-y=15 \rightarrow (1)$$

$$\frac{Y-5}{4}+\frac{y}{4}+\frac{y-10-x}{1}=0$$

$$\Rightarrow 6y-yx=45 \rightarrow (2)$$

From (1), (2) we get



$$Y=\frac{15}{2} w\, x=0$$

$$\therefore I_1=\frac{y-10}{1}=\frac{15}{2}-10=\frac{-5}{2}=2.5$$

CHEMISTRY

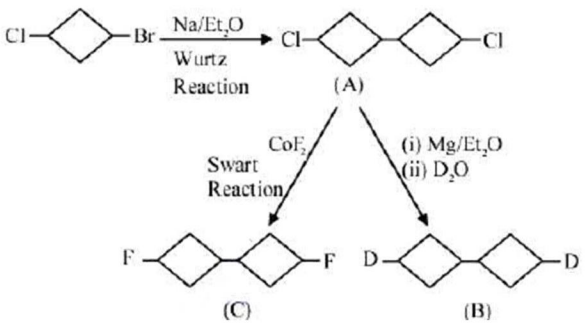
26. $Cu^{+} \rightarrow Cu^{2+}+Cu$

cu^{+} under goes disproportionation due to high hydration energy of Cu^{2+}

$cu(II)$ is more stable than $Cu(I)$ because hydration energy of Cu^{2+} ion compensate $IE2$ of Cu.

27. The chemical formula of compound present in tooth enamel is $[3Ca_3(PO_4)_2.CaF_2]$.

28.



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29. Molar conductivity of weak electrolyte (A) increase sharply than that of strong Electrolyte (B) on dilution

30. Stability of the given ionic species will be decided by the fact that + ve charge on more

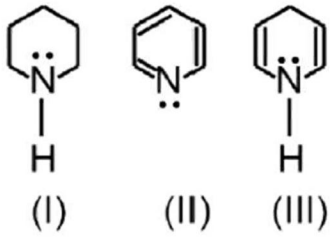
electronegative element will be less stable and as the distance between opposite charges increases, stability decrease, Therefore, the correct stability order is neutral species (I)>(II)>(III) charge separated species.

31. Oxygen can form $2p\pi - 2p\pi$ multiple bond with itself due to its small size while Sulphur cannot form multiple bond with itself as $3p\pi - 3p\pi$ bond will be unstable due to large size of Sulphur, but Sulphur can form multiple bond with small size atom like C and N.

32. Oxidation state is based on electro negativity

33. Total no .of particles in $K_2Cr_2O_7 = 0.03$ while that in $CuSO_4 = 0.1$. So the solution moves from lesser concentration side X to more concentrated side Y. So $CuSO_4$ solution becomes less concentrated.

34.



(I) (II) (III)

In case of III the lone pair of N is delocalized by resonance hence III is weakest. In case of I and II the substituents of I are electron loosing due to their +I effect hence I is strongest nitrogenous base/ Lewis base/base. Piperidine (I)>pyridine(II)>pyrrole (III)

35. The order of percentage of enol forms of different compounds is, III > V > IV > I > II

36. Compounds with more electronegativity difference between hydrogen & oxygen than central atom & oxygen tend to behave as acids.

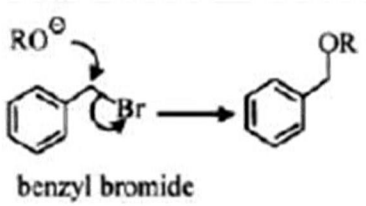
37. S_N2 reaction order of reactivity..

Assertion (A) true, Reason (R) true, ® is the correct explanation of (A)

The benzyl group acts in much the same way using the π - system of the benzene

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ring for conjugation with the P-orbital in the transition state.



benzyl bromide

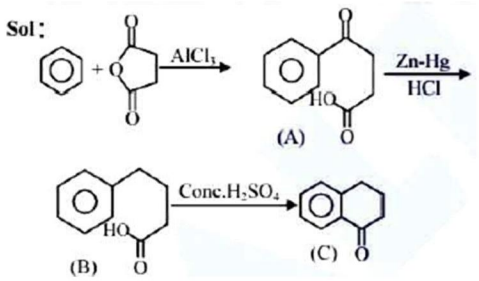
38. Acidic strength order : $B > D > C > A > E$

Correct pKa order : $B < D < C < A < E$

All options are incorrect

Acidic strength $\propto \frac{1}{p^{ka}}$

39.



40. Primary standard should not be hygroscopic like $NaOH$ and should be readily soluble in water $KMnO_4$ is not readily soluble.

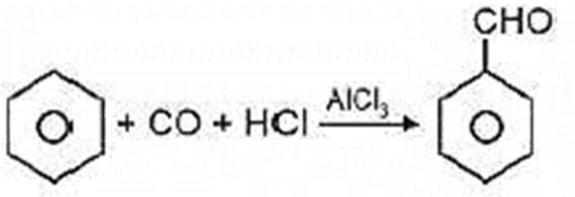
41. *cis* -1

Trans-I (No. of geometrical isomers =2);

No. of stereo isomers =3(cis-isomer exhibit optical activity)

Hybridization is $sp^3d^2 \Rightarrow Geometry$ =octahedral

42. This is gattermann-Koch reaction



43.

Species	Hybridisation
BF_3	SP^2
NO_2^-	SP^2
H_2O	SP^3
NH_2^-	SP^3

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44. Organic compounds are purified based on their nature and impurity present in it, the solubility and density are different from organic compound and on the basis of theses difference compounds are purified. Shape of compounds does not play role in purification

45. System $(n+1)$ value

A	$4+1=5$
B	$4+2=6$
C	$3+1=4$
D	$3+2=5$
E	$4+0=4$

$C < E < D < A < B$ is the energy order as per Aufbau principle?

46. Nucleophiles which have more than one different points of linkage (i.e. electron rich nucleophilic centers are called as ambient nucleophiles

Examples are $CN^-; SCN^-; NO_2^-$

(1) $CN^- \rightarrow C-$ or N - atom can donate e^- pair

(2) $SCN^- \rightarrow S-$ or N - atom can donate e^- pair

(3) $NO_2^- \rightarrow N$ - or O - atom can donate e^- pair

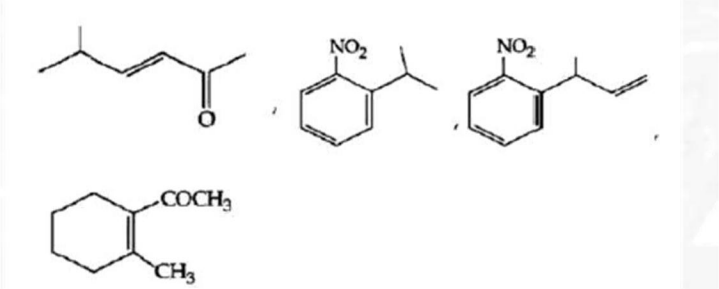
47. Due to higher SRP value of $Cr_2O_7^{-2}$ than Fe^{+3}, Ni^{+2} and $Ag^+, Cr_2O^{-2}{}_7$ can oxidize these three metals Fe, Ni and Ag.

48. Arginine, Phenylalanine, Histidine, Valine, given are essential amino acids

49. $Lu^{3+} = [Xe]4f^{14}5d^06s^0$

Due to completely filled $4f$ subshell or absence of unpaired electron Lu^{3+} is colourless. $La^{3+} = [Xe]4f^0$ noble gas configuration

50. The compounds show inductive, mesomeric as well as hyper conjugation effects are ….



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51. Area of parallelogram with adjacent sides $\overline{OA}$ and $\overline{OC}$ is $|\overline{OA}\times\overline{OC}|=15\Rightarrow|\overline{a}\times\overline{b}|=\frac{5}{2}$

Area of quadrilateral OABC $\frac{1}{2}|\overline{OB}\times\overline{AC}|=\frac{1}{2}|(6\overline{a}+5\overline{b})\times(3\overline{b}-2\overline{a})|=\frac{1}{2}|28(\overline{a}\times\overline{b})|$

$=14\times\frac{5}{2}=35$

52. $\frac{dy}{dx}=\frac{x^2+y^2}{5xy}$ (Homogeneous differential equation)

Put $y=vx\Rightarrow\frac{dy}{dx}=v+x\frac{dv}{dx}$

$\therefore v+x\frac{dv}{dx}=\frac{1+v^2}{5v}$

$\frac{dv}{dx}=\frac{1+v^2}{5v}-v=\frac{1-4v^2}{5v}$

$\frac{5v}{1-4v^2}dv=\frac{dx}{x}$

Integration on both sides

$\int\frac{5v}{1-45v^2}dv=\int\frac{dx}{x}$

$\frac{5}{-8}\log|1-4v^2|=\log x+\log c$

$5\log\left(1-\frac{4y^2}{x^2}\right)=-8\log x+\log c$

$\frac{(x^2-4y^2)^5}{10}=\frac{c}{x^8}$

$(x^2-4y^2)^5=cx^2\rightarrow(1)$

Put $x=1,y=0\Rightarrow c=1$

$\therefore(x^2-4y^2)^5=x^2$

53. $x^2(1+x)^{98}+x^3(1+x)^{97}+x^4(1+x)^{96}+-----x^{54}(1+x)^{46}$

$\downarrow_{x^{68}}\qquad\downarrow_{x^{67}}\qquad\downarrow_{x^{66}}\qquad\qquad\downarrow_{x^{16}}$

Coefficient $x^{70}=98C_{68}+97C_{67}+96C_{66}+------47_{C_{17}}+46_{C_{16}}$

$=46_{C_{15}}+46C_{16}+47C_{17}+-----98C_{68}-46C_{15}$

$=99C_{68}-46C_{45}(\therefore n_{Cr}+nC_{r-1}=n+1_{C_r})$

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$P = 68, q = 15$

$p = q = 83$

54. $\frac{x-2}{1} = \frac{y}{-1} = \frac{z-1}{1} = t = L_1$

$\frac{x+1}{1/2} = \frac{y-1}{1/2} = \frac{z+}{1} = s = L_2 dr's = (1,1,2)$

$P(t+2,-t,t+1)$ lies on L_1 and

$Q(s-1,s+1,2s-1)$ lies on L_2

Dr'S of $P\overline{q} = (s-1-t-2, s+1+t, 2s-1-t-1)$ parallel to (3,1,2)

$\frac{s-t-3}{3} = \frac{s+t+1}{1} = \frac{2s-t-2}{2}$

$s-t-3 = 3s+3t+3$

$2s+4t+6 = 0$

$s+2t+2 = 2s-t-2$

$3t = -4$

$t = \frac{-4}{3}$

$P = \left(\frac{-4}{3}+2, \frac{4}{3}, \frac{-4}{3}+1\right) = \left(\frac{2}{3}, \frac{4}{3}, \frac{-1}{3}\right)$

Lines passes through P and parallel to (3,1,2)

$\Rightarrow \frac{x-\frac{2}{3}}{3} = \frac{y-\frac{4}{3}}{1} = \frac{z+\frac{1}{3}}{2}$

Substitute $\left(\frac{-1}{3}, 1, -1\right)$ we get

$\left(\frac{-1}{3}, 1, -1\right)$ exist

55. $|\vec{c}| = \overline{a} - \overline{b} = (\alpha - 5)\overline{i} + \overline{j} - 2\overline{k}$

Area of $\Delta ABC = \frac{1}{2}|\overline{a} \times \overline{b}| = 5\sqrt{6}$

$$\begin{vmatrix} \overline{i} & \overline{j} & \overline{k} \\ \alpha & 4 & 2 \\ 5 & 3 & 4 \end{vmatrix} = 10\sqrt{6}$$

$|10\overline{i} - j(4\alpha - 10) + \overline{k}(3\alpha - 20)| = 10\sqrt{6}$

$100 + (4\alpha - 10)^2 + (3\alpha - 20)^2 = 600$

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$\alpha - 10)^2 + (3\alpha - 20)^2 = 600$

$25\alpha^2 - 200\alpha = 0$

$\alpha = 8$

$\therefore \bar{c} = 3\bar{i} + \bar{j} - 2\bar{k} \Rightarrow |\bar{c}|^2 = 14$

56. $\Delta MAP, \tan\theta = \dfrac{5}{MA}$

$MA = 5\cot\theta$

$\Delta nPB, \tan\theta = \dfrac{NB}{3}$

$NB = 3\tan\theta$

$\therefore OA = 3 + 5\cot\theta, OB = 5 + 3\tan\theta$

Area of $\Delta OAB = \dfrac{1}{2}(3 + 5\cot\theta)(5 + 3\tan\theta)$

$= \dfrac{1}{2}(30 + 9\tan\theta + 25\cot\theta)$

$\geq (30 + 2\sqrt{9\tan\theta \times 25\cot\theta})$

(use AM $\geq$ G.M)

$\geq \dfrac{1}{2}(30 + 2 \times 3 \times 5)$

≥ 30

57. $x^2 + y^2 = 5 \rightarrow (1), y^2 = 4x \rightarrow (2)$

Solving (1)& (2) ; $x^2 + 4x = 5$

$x^2 + 4x = 5$

$(x + 5)(x - 1) = 0$

$x = 1, x \neq -5$

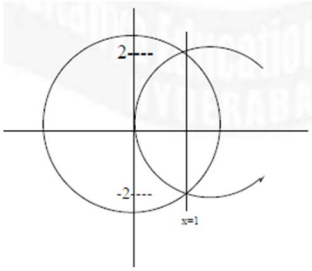
Put of the region bounded by circle and parabola is $A_1 = \int_{-2}^{2} \sqrt{5 - y^2} - \dfrac{Y^2}{4} dy$

$A_1 = 2\int_0^2 \sqrt{(\sqrt{5})^2 - y^2} - \dfrac{y^2}{4} dy$

$A_1 = 2\left[\left(\dfrac{y}{2}\sqrt{5 - y^2} + \dfrac{5}{2}\sin^{-1}\dfrac{y}{\sqrt{5}}\right) - \dfrac{"y^3}{12}\right]_0^2$

$A_1 = 2\left[1 + \dfrac{5}{2}\sin^{-1}\dfrac{2}{\sqrt{5}} - \dfrac{2}{3}\right]$

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$$A_1 = \frac{2}{3} + 5\sin^{-1}\frac{2}{\sqrt{5}}$$

58. $$\frac{1}{d}\left[\frac{d}{1(1+d)} + \frac{d}{(1+d)(1+2d)} + ------ + \frac{d}{(1+9)(1+10d)}\right] = 5$$

$$\frac{1}{d}\left[\frac{(1+d)-1}{1(1+d)} + \frac{(1+2d)-(1+d)}{(1+d)(1+2d)} + ------ + \frac{(1+10d)-(1+9d)}{(1+9d)(1+10d)}\right] = 5$$

$$\frac{1}{d}\left[\frac{1}{1} - \frac{1}{1+d} - \frac{1}{1+2d} + .----- + \frac{1}{1+9d} - \frac{1}{1+10d}\right] = 5$$

$$\frac{1}{d}\left[\frac{1}{1} - \frac{1}{1+10d}\right] = 5$$

$$\frac{1}{d}\left[\frac{10d}{1+10d}\right] = 5 \Rightarrow 2 = 1+10d$$

$$10d = 1$$

$$d = \frac{1}{10}$$

$$50d = 5$$

59. $$\frac{x-3}{4} = \frac{y+7}{-11} = \frac{z-1}{5}, \frac{x-5}{3} = \frac{y-9}{-6} = \frac{z+2}{1}$$

$$\bar{a} = (3,-7,1)$$

$$\bar{b} = (4,-11,5)$$

$$\bar{c} = (5,9,-2)$$

$$\bar{c} = (5,9,-2)$$

$$\bar{b} \times \bar{d} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 4 & -11 & 5 \\ 3 & -6 & 1 \end{vmatrix} = 19\bar{j} + 11\bar{j} + +9\bar{k}$$

$$\bar{b} \times \bar{d} = \sqrt{19^2 + 11^2 + 9^2} = \sqrt{563}$$

$$[\bar{a} - \bar{c}\bar{b}\bar{d}] - = \begin{vmatrix} -2 & -16 & 3 \\ 4 & -11 & 5 \\ 3 & -6 & 1 \end{vmatrix} = -2(19) + 16(11) + 3(9)$$

$$= -185$$

$$\text{Shortest distance} = \frac{\left|[\bar{a} - \bar{c}\ \bar{b}\ \bar{d}]\right|}{|\bar{b} \times \bar{d}|} = \frac{185}{\sqrt{563}}$$

60. $$a = fog(10) = f(g(10)) = f(10) = 109$$

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$b = gof(3) = g(f(3)) = g(18) = \frac{18}{9} = 2$

$\frac{x^2}{109} + \frac{y^2}{2} = 1$

$e = \sqrt{\frac{109-2}{109}} = \sqrt{\frac{107}{109}}; l = \frac{2b^2}{a} = \frac{2(2)}{\sqrt{109}}$

$8e^2 + l^2 = 8 \times \frac{107}{109} + \frac{16}{109} = \frac{8(109)}{109} = 8$

61. $f(x) = ax^2 + bx^2 + cx + 41$

$f'(x) = 3ax^2 + 2bx + c$

$f''(x) = 6ax + 2b$

$f(1) = a + b + c + 41 = 40$

$a + b + c = -1 \rightarrow (1)$

$f'(1) = 3a + 2b + c = 2 \rightarrow (2)$

$f''(1) = 6a + 2b = 4$

After solving (1), (2) & (3)

We get $a = -1, b = 5, c = -5$

$a^2 + b^2 + c^2 = 51$

62. Given 's' be a circle it passes through (2,0)

$centr(h.k)$

$(xc + yc)$ be the POI

Of $3x + 5y = 1 ------, (1)$

$(2 + c)x + 5c^2 y = 1$

h= $\lim_{c \to \infty} x_c; k = \lim_{c \to 1} y_c$

Solving (1) & (2)

$(2 + c)x + 5c^2 y = 1$

$3c^2 x + 5c^2 y = c^2$

$(2 + c - 3c^2)x = 1 - c^2$

$= -(-2\sqrt{2}) \ -1(-1)$

Sub (3) in (1)

$y_c = \frac{c-1}{5(2 + c - 3c^2)}$

$h = \lim_{c \to 1} x_c = \lim_{c \to 1} \left(\frac{1 - c^2}{2 + c - 3c^2} \right) \left(\therefore \frac{0}{0} form \right)$

$h = \lim_{c \to 1} \left(\frac{-2c}{1 - 6c} \right)$

$h = \frac{2}{5}$

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$$k = \lim_{c\to 1} y_c = \lim_{c\to 1}\left(\frac{c-1}{5(2+c-3c^2)}\right)$$

$$k = \left(\frac{-1}{25}\right)$$

$$(h,k) = \left(\frac{2}{5},\frac{-1}{25}\right)$$

$$r = \sqrt{\left(2-\frac{2}{5}\right)^2 + \left(\frac{1}{25}\right)^2}$$

$$r^2 = \frac{1601}{625}$$

Equation of circle $= \left(x-\frac{2}{5}\right)^2 + \left(y+\frac{1}{25}\right)^2 = \frac{1601}{625}$

$$x^2 + \frac{4}{25} - \frac{4x}{5} + y^2 + \frac{1}{625} + \frac{2y}{25} = \frac{1601}{625}$$

$$25x^2 + 25y^2 - 20x + 2y - 60 = 0$$

63. Given

$$de = 2y\frac{dy}{dx} + 3 = 5\frac{dy}{dx}$$

It passes through (0,1)

$$(2y-5)\frac{dy}{dx} = -3$$

$$(2y-5)dy = -3dx$$

$$2\frac{y^2}{2} - 5y = -3x + c$$

$$y^2 - 5y + 3x = c$$

$$c = -4$$

$$y^2 - 5y + 3x = -4$$

$$y^2 - 5y + 3x = -4$$

$$y^2 - 5y + \frac{25}{4} - \frac{-25}{4} = -4 - 3x$$

$$\left(y-\frac{5}{2}\right)^2 = -3x + \frac{9}{4}$$

$$\left(y-\frac{5}{2}\right)^2 = -3\left(x-\frac{3}{4}\right)$$

Vertex=$\left(\frac{3}{4},\frac{5}{2}\right)$

It lies on $2x+3y=9$

64. T be the image of (1,2)

Slope of QR=slope QT

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$$\frac{3}{4-x}=\frac{2}{x-1}$$

$$3x-3=8-2x$$

$$5x=11$$

$$x=\frac{11}{5}$$

$$Q\left(\frac{11}{5}10\right)$$

Given PQRS is parallelogram

Midpoint of PR=Midpoint of QS

$$\left(\frac{5}{2},\frac{5}{2}\right)=\left(\frac{h+\frac{11}{5}}{2},\frac{k}{2}\right)$$

$$\therefore \frac{k}{2}=\frac{5}{2}\Rightarrow k=5$$

$$\frac{5}{2}=\frac{n+\frac{11}{5}}{2}$$

$$n=\frac{14}{5}\therefore hk^2=\frac{14}{5}\times 25$$

$$hk^2=70$$

65.

Age (xi)	No. of students fi	Cumulative frequency	|xi-medium|	$f_i=\|x_1-m\|$
15	5	5	3	15
16	8	13	2	16
17	5	18	1	5
18	12	30	0	0
19	X	30+x	1	X
20	Y	30+x+y	2	2y
	N=40			36+x+2y

$$30+x+y=40; N=40$$

$$x+y=10\rightarrow(1)\frac{N}{2}=20$$

Medium=18

$$\text{M.D}=\frac{1}{N}\sum fi\,|x_i-m|=\frac{36+x+2y}{40}=1.25$$

$$\Rightarrow x+2y=14\rightarrow(2)$$

Solving (1)&(2) we get $x=6, y=4$

$$4x+5y=24+20=44$$

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66. $f(x)=\sin^{-1}\left(\frac{x-1}{2x+3}\right)$

$-1\le\frac{x-1}{2x+3}\le 1$

$-2x-3\le x-1$

$-3x-2\le 0$

$-3x\le 2$

$x\le\frac{-2}{3}$

$x-1\le 2x+3$

$0\le x+4$

$-4\le x$

Domain $=(-\infty,-4]U\left[\frac{-2}{3},\infty\right]$

$\Rightarrow R-\left(-4.\frac{-2}{3}\right)$

$\alpha=-4'\beta=\frac{-2}{3}$

$12\alpha\beta=12\times-4\times\frac{2}{3}$

$12\alpha\beta=32$

67. $3x+2y+\lambda z=3$

$7x+11y-9z=2$

$97x+155y-189z=\mu$

The given system of equation has infinitely many solutions

$\therefore\Delta=0;\Delta_1=0;\Delta_2=0$

$\Delta_3=0$

$\Delta\begin{vmatrix}3 & 5 & \lambda\\ 7 & 11 & -9\\ 97 & 155 & -189\end{vmatrix}=0$

$\Rightarrow\lambda=-11$

$\Delta_1=0$

$\begin{vmatrix}3 & 5 & -11\\ 2 & 11 & -9\\ \mu & 155 & -189\end{vmatrix}=0$

$76\mu=3572$

$\mu=47$

$\mu+2\lambda=47+2(-11)$

$=47-22$

$\mu=25$

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68. $\int \frac{2-\tan x}{3+\tan x}dx = \frac{1}{2}(\alpha x + \log_e |\beta \sin x + \gamma \cos x|) + c$

$$\int \frac{\frac{2-\sin x}{\cos x}}{3+\frac{\sin x}{\cos x}} = \int \frac{2\cos x - \sin x}{3\cos x + \sin x}$$

$2\cos x - \sin x = A(3\cos x + \sin x + B(\cos x = 3\sin x))$

By comparing Coefficient

$3A + 2B = 2 \rightarrow (1)$

$A - 3B = -1 \rightarrow (2)$

Solving (1) & (2)

$a - 3B = -1$

$9a + 3b = 6$

$10a = 5$

$A = \frac{1}{2}$

$B = \frac{1}{2}$

$a = 1' \beta = 1'8 = 3$

$\alpha + \frac{8}{\beta} = 1 + 3 = 4$

69. Given

α, β ae the roots of equation $x^2 + 2\sqrt{2x} - 1 = 0$

Then the quadratic equation whose roots are

$\alpha^2 + \beta^4 = \frac{1}{10}(\alpha^6 + \beta^6)$

$\alpha^2 + \beta^4 = (\alpha + \beta)^2 - 2\alpha\beta$

$\frac{\alpha^6 + \beta^6}{10} = 97$

$\alpha^2 + \beta^2 = 8 + 2$

$\alpha^2 + \beta^2 = 10$

$\alpha^4 + \beta^4 = (\alpha^2 + \beta^2) - 2(\alpha\beta)^2$

$= 100 - 2(1) = 98$

$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

$= -16\sqrt{2} - 3(-1)(-2\sqrt{2})$

$\alpha^6 + \beta^6 = (\alpha^3 + \beta^3)^2 - 2(\alpha\beta)^3 \quad \frac{1}{10}(\alpha^2 + \beta^6) \Rightarrow x^2 - (97 + 98)x + 98 \times 97 = 0$

$= 484 \times 2 + 2$

$\alpha^6 + \beta^6 = 968 + 2$

$968 + 2 = 970$

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$$\frac{\alpha^6+\beta^6}{10}=97=$$

Quadratic Equation whose roots are $(\alpha^4+\beta^4)$ and

$$\frac{1}{10}(\alpha^2+\beta^6)\Rightarrow x^2-(97+98)x+98\times 97=0$$

$$x^2-195x+9506=0$$

70. $|COS\theta\cos(60-\theta)\cos(60+\theta)\le\frac{1}{8}$

$$|\frac{1}{4}\cos 3\theta|\le\frac{1}{8}$$

$$\frac{-1}{8}\le\frac{1}{4}\cos 3\theta\le\frac{1}{8}$$

$$\frac{-1}{2}\le\cos 3\theta\le\frac{1}{2}$$

From above max value of $\cos\theta$ is $\frac{1}{2}$

$$\therefore\cos 3\theta=\frac{1}{2}$$

$$3\theta=2\pi\pm\frac{\pi}{3}$$

$$\theta=\frac{2n\pi}{3.}\pm\frac{\pi}{9}$$

$$n=0\Rightarrow\theta=\frac{\pi}{9}$$

$$n=1\Rightarrow\theta=\frac{7\pi}{9},\frac{\pi}{9}$$

$$n=2\Rightarrow\theta=\frac{13\pi}{9},\frac{11\pi}{9}$$

$$n=3\Rightarrow\theta=\frac{17\pi}{9}$$

Sum of all values of θ in $(0,2\pi)$

$$=\frac{\pi}{9}+\frac{7\pi}{9}+\frac{5\pi}{9}+\frac{13\pi}{9}+\frac{11\pi}{9}+\frac{17\pi}{9}=\frac{54\pi}{9}=6\pi$$

71. $A=\{2,3,6,7\}$

$B=\{4,5,6,8\}$

$(a,b)R(a_1,b_2)\Rightarrow a_1+a_\alpha=b_1+b_2$

$a_1,a_2\in A$ and $b_1+b_2\in B$

$4\le a_1+a_2\le 14, 8\le b_1+b_2\le 16$

Since $a_1+a_2=b_1+b_2\Rightarrow$ consider 8,9-----14

$((a_1,a_2)\quad (b_1,b_2)$

$8\to(2,6)\qquad(4,4)\to 2$

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$9 \to (3,6)(2,7) \quad (4,5) \to 8$

$10 \to (3,7) \quad (4,6)(5,5) \to 6$

$11 \to (5,6) \to 0$

$12 \to (6,6) \quad (4,8)(6,6) \to 4$

$14 \to (7,7) \quad (6,8) \to 1$

Total elements in the relation are 25

72. $\lim\limits_{x \to \frac{\pi}{2}-} \left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}} = \lim\limits_{x \to \frac{\pi}{2}} \left(\frac{8}{7}\right)^{\tan 8x \cot 7x}$

$\lim\limits_{x \to \frac{\pi}{2}+} (1-\cot x)^{\frac{-b}{a}\tan x} = \left(\frac{8}{7}\right) = 1$

$|3m+2n|$

$a - 81 = e^{\frac{b}{a}}$

$a = 9, \frac{b}{a} = 0 \Rightarrow b = 0$

$a^2 + b^2 = 9^2 + 0 = 81$

73. Let $z = x + iy, x, y \in z$

$|z-1| \le 1 \Rightarrow (x-1)^2 + y^2 \le 1 \to (1)$

$|z-5| \le |z-5i|$

$(x-5)^2 + y^2 \le x^2 + (y-5)^2$

$\Rightarrow y \le x \to (2)$

Solving $(1)\,\&\,(2)$ we get $x = 1$

The integral points in the region are $(1,1)(1,-1)(1,0)$ and $(2,0)$

Square of modules of the elements

74. Given $\Rightarrow 3^3 2^6 \,|\, adj \,|\, A \,|^2 \; adj(A) | = 3^{-13} 2^{-10}$

$|3adj(2adj(|\, A \,|\, A))| = 3^{-13} 2^{-10}$

$3^3 \,|\, adj(2adj(|\, A \,|\, A))| = 3^{-13} 2^{-10}$

$(\because |KA| = K^n \,|\, A \,|\, adj(KA) = K^{n-1} adjA)$

$\Rightarrow 3^3 \,|\, 2^2 adjadj(|\, A \,|\, A)| = 3^{13} 2^{-10}$

$\Rightarrow 3^3 2^6 \,|\, adj \,|\, A \,|^2 \; adj(A)| = 3^{-13} 2^{-10}$

$\left| |\, A \,|^4 \; adjadj(A)| = 3^{-16} 2^{16} \right|$

$|\, A \,|^{12} \; adjadj(A) = 3^{-16} 2^{-16}$

$|\, A \,|^{12} |\, A \,|^{(3-1)^2} = 3^{-16} 2^{-16}$

$|\, A \,|^{16} = 3^{-16} 2^{-16}$

$|\, A | = 3^{-1} 2^{-1}$

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Now $|3adj(2A)| = 3^3 |adj(2A)|$

$= 3^3 |2^2 adj(A)|$

$3^3 2^6 |A|^2 = 3^3 2^6 3^{-2} 2^{-2} = 3.2^4$

$m = 4, n = 1$

$|3m + 2n| = |12 + 2| = 14$

75. a, b, c

$ax^2 bx + c = 0$ has real roots

$\Rightarrow b^2 - 4ac \geq 0$

$b^2 \geq 4ac$ If

$n(s) = 4^3 = 64$

Suppose

If $b = 1 \Rightarrow 1 \geq 4ac$ no value of a,c exist

If $b = 2 \Rightarrow 4 \geq 4ac$

$1 \geq ac$

$a = 1, c = 1$

If $b = 3 \Rightarrow p \geq 4ac$

$a = 1, c = 1$ and $a = 2, c = 1, anda = 1, c = 2$

If $b = 4 \Rightarrow 16 \geq 4ac$

$a = 1, c = 1and$

$a = 1, c = 2$ and

$a = 3, c = 1$ and

$a = 1, c = 3$ and

$a = 2, c = 2$ and

$a = 4, c = 1$ and

$a = 1, c = 4$

$P(A) = \frac{n(A)}{n(S)} = \frac{12}{64} = \frac{3}{16} = \frac{m}{n}$

$m + n = 3 + 16 = 19$

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